## Appendix A: Interview Protocol

*This appendix summarizes the structured interview protocol used in this study. The questions were organized into seven thematic categories designed to capture participants' experiences, practices, and perspectives on log analysis, parsing, and anomaly detection.*

*Questions marked with an asterisk (*) were asked in every interview, while the remaining questions were posed selectively depending on participants' responses and professional background.*

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Thanks for taking the time to meet with me today. We are investigating the practices for log analysis. During the interview, I'd like to ask you a few questions about your background, and then we will discuss your experience.

Do you mind if I record this discussion so that I can review it later? In any data collected, or in reports or papers that are published, you will not be identified by name. Please be careful not to discuss any sensitive information about the company you work for. If you do mention any, we will do our best to remove them from our transcripts, but it would be better if you did not mention such sensitive information at all.

**Participant Background and Professional Experience**

a) *How many years of experience do you have in software development, operations, or system maintenance?
b) *Could you briefly describe the most recent software project you have worked on, including its primary function and technical architecture?
c) *What specific role did you play in this project, and what were your main responsibilities related to system monitoring or log management?
d) *Have you been involved in and received any specific training related to log analysis, system monitoring, or anomaly detection?
e) What types of systems or platforms have you primarily worked with during your career (e.g., web services, IoT, financial, or industrial)?

**Log-related Experience**

a) *How large is your current development or operations team, and how is the responsibility for log analysis distributed among team members?
b) *How frequently do you or your team review logs as part of regular maintenance or debugging activities?
c) *When a system anomaly is detected, how does your organization coordinate between developers, DevOps engineers, and managers to address it?
d) *Are there formal protocols or documentation practices for recording and sharing anomaly cases across the team?
e) How are priorities determined when multiple alerts or anomalies occur simultaneously?

**Log Management Tools and Practices**

a) *Have you or your team used or implemented any automated tools or frameworks for log collection, parsing, or anomaly detection? If so, which ones and why?
b) *What factors influence your team's choice of log management tools (e.g., scalability, interpretability, cost, integration)? If applicable, does your team develop or customize any in-house tools for internal use?

**Log Characteristics and Parsing Challenges**

a) *Can you describe the main types of logs generated by the systems you currently develop or maintain?
b) *What are the most common normal events and anomaly types you encounter during system operation?
c) *Do you or your team currently adopt any form of log parsing during data preprocessing or monitoring workflows? If yes, have you observed cases where automated parsing tools misinterpreted or discarded important information? Could you provide an example?
d) *From your perspective, what characteristics make a log message particularly "difficult" to parse or classify? If your team currently performs log parsing during preprocessing or monitoring (as mentioned earlier), could you describe specific situations or message types that were especially challenging to handle?
e) In your experience, what are the most challenging aspects of log parsing or preprocessing?

**Anomaly Detection and Diagnostic Practices**

a) *How do you or your team currently identify anomalies in system logs? Are these processes automated, manual, or hybrid?
b) *How frequently do you encounter rare or unexpected anomalies, and how do you typically handle them?
c) *Can you recall a situation where a log anomaly led to a significant system issue or downtime? How was it identified and resolved?
d) *Could you share several examples of log entries that you consider typical representations of normal or anomalous behavior?
e) What difficulties do you face when distinguishing between false positives and true anomalies in log monitoring?

**Future Perspectives and AI Integration**

a) *In your opinion, how could AI- or LLM-based tools improve current log analysis workflows in your organization?
b) *If an AI-based tool could automatically detect and summarize anomalies, what kind of explanations or features would you expect it to provide?
c) What potential risks or limitations do you anticipate in applying deep learning or prompt-based LLM approaches to log parsing, anomaly detection, and other log analysis tasks?